

Pre-Feasibility Report for Turamdih Mines Augmentation

1.0 EXECUTIVE SUMMARY:

The salient features of the project include:

Proposal	Augmentation of production capacity of Turamdih mines from 0.45 million TPA to 0.75 million TPA without change in mine lease area.
Location of Project	Uranium Corporation of India Ltd. Turamdih Mines PO:- Sundernagar, Dist: East Singhbhum, Jharkhand-832107
Latitude	22 ⁰ 43'9.6"N to 22 ⁰ 44'11.2"N
Longitude	86 ⁰ 10'34.1"E to 86 ⁰ 11'44.7" E
Land Ownership of Lease Area	Total mine lease area will be 238.344 ha. At present mine lease over 225.487 ha (557.18 acres) has been granted. Application for additional mine lease over 12.857 ha has been made.
Method of Work	Mechanized underground working by Horizontal Cut and Fill (HCF) method.
Mineral Reserve	7.5 million tones w.e.f. 10.10.2020
Production capacity	Existing: 0.45 million TPA Proposed: 0.75 million TPA
Waste Generation	~1,50,000 t/yr
Waste disposal	About 100,000 ton of waste of new decline development and vertical shaft shall be dumped at the designated waste dump area inside the mine lease. The waste about 150000 ton/yr shall be generated and be used as backfilled material in underground at excavated voids of stope.
Mineral Transport	By trucks and conveyor to adjacent Turamdih Mill and nearby proposed ore processing plant at Banduhurang.
Mineral Processing	At Turamdih Ore Processing Plant and nearby proposed ore processing plant at Banduhurang. The uranium is extracted as Magnesium di urinate / uranium peroxide by a hydro-metallurgical process. The MDU is dispatched to Nuclear Fuel Complex Hyderabad for further processing.
Working Regime	3 shifts per day; 300 working days per year.
Life of Mine	8 years w.e.f 10.10.2020
Peak Water Demand	Water requirement will be 3430 m ³ /day. Out of

	which 3170 m ³ /day will be met by mine discharge water and 260 m ³ /day will be tapped from the existing water supply system.
Source of water	Industrial: Treated mine discharge Potable: Kharkhai River a tributary of Subarnarekha River.
Power Demand	The present power consumption is 2.0 MW. The power demand is not expected to increase to the level of 5 MW.
Source of power	Jharkhand State Electricity Board's Ichra Sub-station
Man Power	Present: 850 and additional 500 will be required
Explosive Consumption	The present explosive consumption is 0.5 tons per day, which is expected to increase to 1.5 tons per day.
Fuel Consumption	The present HSD consumption is 0.57 tons per day which is expected to increase to 1.4 tons per day.
Infrastructure	New production shaft, Decline, Ventilation Shaft, workshop, Mine Vehicle, Crusher and related equipment, Pumping, Ventilation Fan, Winding Equipment etc
Proposed Investment	Rs 300 crore
Production Cost	Classified information
CSR Budget	2% of net profit

The proposed expansion project will improve the supply of uranium ore which will provide fuel for India's expanding nuclear energy programme. The expansion project will generate both direct and indirect employment. The project will pave the way for peripheral development in a predominantly tribal area.

2.0 INTRODUCTION OF THE PROJECT / BACKGROUND INFORMATION :

2.1 IDENTIFICATION OF PROJECT AND PROJECT PROPONENT:

Identification of project

Turamdih Uranium deposit is located in the Dhalbhum subdivision of East Singhbhum district of Jharkhand state. The mine site is located at a Latitude 22° 43' 17" N to 22° 44' 36" N and longitude 86° 10' 39" E to 86° 11' 42" E, which is covered by Survey of India toposheet no. 73J/2 on R.F. 1:50,000 (1978).

The mine is presently working with a production capacity of 0.45 million TPA. It is planned to increase the same to 0.75 million TPA. Additional mining lease over 12.857 ha area have been applied.

Project proponent

The project proponent is Uranium Corporation of India Limited (UCIL) Jaduguda, a Government of India Enterprise under the administrative control of department of Atomic Energy, Govt. of India. UCIL's existing mining operations are spread over Jharkhand and Andhra Pradesh. UCIL is operating six underground mines (Jaduguda, Bhatin, Bagjata, Turamdih, and Mohuldih in Jharkhand and Tummalapalle in Andhra Pradesh) and one open cast mine (Banduhurang in Jharkhand). UCIL also operates uranium ore processing plants at Jaduguda and Turamdih in Jharkhand and Tummalapalle in Andhra Pradesh to extract the uranium present in the ore.. The uranium ore is extracted as magnesium-diuranate / uranium peroxide at the Jaduguda and Turamdih plants and as sodium-di-uranate in the Tummalapalle plant also known as "Yellow Cake". The yellow cake is dispatched to Nuclear Fuel Complex, Hyderabad for further processing.

2.2 BRIEF INFORMATION OF THE PROJECT:

Turamdih Mine is an underground mine being operated by Uranium Corporation of India Ltd. The mine is located in the Dhalbhum subdivision of East Singhbhum district of Jharkhand state. The present lease area of Turamdih Mine had been granted for mining operation over an area of 225.487 ha (557.18 ac) under village of Nandup, Banduhurang and Purihasa by Deputy Secretary (Mines) vide their letter no. 566/M.RANCHI dated 26.04.2008. The lease has been granted during August 2008 for a period of 30 years with option of renewal for a further period of 20 years. In exercise of the powers conferred by section 11B of MMRD Act 1957, the Central Government has notified the Atomic Mineral Concession Rule, 2016 (AMCR, 2016) on 11th July 2016. As per the Rule 11 of AMCR 2016, "The mining leases granted under these rules, shall be granted for a period until the entire reserves of such minerals in the mine is exhausted: Provided that an existing mining lease as on date of coming into force of these rules granted to a Government company for extraction of prescribed substances including uranium or thorium shall be deemed to have been granted for a period until the entire reserves of such minerals in the mine is exhausted". According to the above rule, the mining lease of Turamdih mine has been extended for a period until the entire reserves is exhausted. Additional mining lease over 12.857 ha area in mauza Nandup have been applied to Govt. of Jharkhand in 2006. Forestry clearance for 12.595 ha (out of 12.857 ha) has been applied in 2006 to the Nodal office , Govt. of Jharkhand.

2.3 NEED FOR THE PROJECT AND ITS IMPORTANCE TO THE COUNTRY OR REGION :

Govt. of India has planned to increase nuclear power generation from 4120 MWe at present to 20,000 MWe by the year 2020. This will lead to increased demand for Uranium. In order to meet the increased demand for uranium, UCIL has planned to increase the production from its existing mines besides developing new mines. Expansion of Narwapahar Mine is part of this programme to meet the increasing demand for uranium for India's nuclear power industry. India's uranium resources are scarce, only about 0.8% of the world's uranium deposits. However India contains 20 – 25% of the world's thorium deposits. India is developing the technology to utilise thorium in its nuclear power programme in a three stage programme. The first two stages require uranium. India has to exploit the existing uranium deposits. The proposed expansion programme will increase the availability of nuclear fuel for the 1st two stages of the country's nuclear power programme. The proposed expansion project will also generate employment for local inhabitants in a predominantly tribal area. UCIL will spend part of the profits from the expanded mine for peripheral development which will benefit local villagers.

2.4 DEMAND AND SUPPLY GAP :

Demand of fuel for nuclear reactor is not fully met by indigenous uranium. At present with 6780 MW of installed nuclear power capacity, about 32% of fuel requirement is met by imported uranium. With addition of more nuclear power reactors, demand and supply gap of uranium is likely to increase.

2.5 IMPORT VS INDIGENOUS PRODUCTION :

Refer clause 2.4 above.

2.6 EXPORT POSSIBILITIES :

There is no possibility of any export of uranium ore or concentrate from India.

2.7 DOMESTIC / EXPORT MARKET:

There is no possibility of domestic and export market for uranium ore supply.

2.8 EMPLOYMENT GENERATION:

The mine employs 800 persons consisting of executives, supervisors and workers. Additional 550 persons shall be deployed.

3.0 PROJECT DESCRIPTION

3.1 TYPE OF PROJECT INCLUDING INTERLINKED AND INTERDEPENDENT PROJECT

The present project envisages capacity enhancement of the existing underground mine without increase in mine lease area. The ore from Turamdih is sent to the nearby Ore Processing Plant at Turamdih which processes the ore from Turamdih mines, Banduhurang open cast mines and Mohuldih mines. The Plant has environmental clearance. The Department of Environment, Forests & Wildlife have granted environmental clearance for mine for 1500 tpd and ore processing plant vide letter no J-11015118/85-En.5/IA dated 22nd September 1987 and letter No. J-11011/9/86-IA dated 30th April 1987 for 3000 tpd respectively. Application has been made to increase the production capacity of processing plant from 3000 tpd to 4500 tpd. The Environmental Appraisal Committee has recommended for environmental clearance on the 2nd Meeting of EAC held on 21st December 2012.

3.2 LOCATION:

Turamdih uranium deposit is located in the Dhalbhum subdivision of East Singhbhum district of Jharkhand state. The distance from the Tatanagar railway station to Turamdih is about 6 km. The deposit is covered under survey of India toposheet no 73 J/2 bounded between latitudes 22° 43' 17" N to 22° 44' 36" N and longitudes 86° 10' 39" E to 86° 11' 42" E. The deposit is well connected by metal roads with the nearest city Jamshedpur. The head office of UCIL is located at Jaduguda at about 25 km from Turamdih.

3.3 DETAILS OF ALTERNATE SITE:

Since the project envisages capacity increase of an existing underground mine and mining is a site specific activity guided by deposit geology, the question of any alternate site does not arise.

3.4 SIZE AND MAGNITUDE OF OPERATION:

The mine lease is spread over 225.487 ha. The mine's present capacity is 0.45 million TPA of ROM ore. The present proposal envisages increasing the ROM production to 0.75 million TPA, without any increase in mine lease area.

3.5 MINE DESCRIPTION

3.5.1 Geology and Exploration

Turamdih uranium deposit falls in the west-central part of the Singhbhum Shear Zone. Feldspathic-schist, chlorite sericite schist with magnetite, apatite and quartzite are the host rocks for uranium mineralization, which belong to the Dalbhum Formation of Proterozoic age (1.48-1.58 Ga; Misra and Johnson, 2005). Uranium in the form of mineral Uraninite is disseminated within the schistose rocks and appears to follow the foliation planes. This mineralized (Cu-U) stretch is a part of the 160 km arcuate belt known as Singhbhum Shear Zone, outstanding for its geological attributes and extensive mining activities.

Chlorite-schist and feldspathic-quartzite-schist are the two major predominant lithounits hosting uranium mineralization at Turamdih. The overlying quartzite horizon with NW-SE elongation is largely unmineralised. Quartzite and sericite schists are hanging wall and footwall marker horizons of uranium mineralization respectively. Uranium mineralization represented by uraninite is generally associated with the oxides like magnetite and Ilmenite. Local geological succession of the east Singhbhum district is given below:

Alluvium
Laterite
Tertiary Grits and Gravels
Newer dolerites
Soda granites/pyroxene granites
Dalma, dhanjori lava
Dalbhum formation
Chaibasa formation
Singhbhum granite
Iron ore series –phyllites and schists
Older metamorphic

3.5.2 Mineral Reserve

Total mineral reserve is 7.5 MT as on 09.10.2017.

3.5.3 Method of Mining

Status of Mine

Turamdih Mine is being operated using two numbers of mine entries namely 8 degree Decline and a vertical Shaft of 5 meter diameter. The mine is producing nearly 1000 tpd at present. Two numbers of ventilation shaft namely Eastern and western ventilation shaft respectively are being used to ventilate the mine. The mine

has the following working levels, viz. 60-m bgl (1st level: 120 mRL), 100-m bgl (2nd level: 80 mRL), 140 m bgl (3rd level: 40 mRL), 180 m bgl (4th level: Zero mRL), 200 m bgl (5th level /Crushing Station: -20 mRL), 230 m bgl (6th level /Hoisting Station: -50 mRL). The main shaft bottom is at a depth of 260 m depth (-80 mRL). The main shaft is connected to all the working levels. The 8-degree Decline has been developed up to 260 m level. The Eastern, Central & the Western ventilation shafts (3.5 m finished diameter) are connected to 1st level. Due to enhanced requirement of uranium for the country it has been proposed to achieve ore production capacity up to 2500 tpd. For enhancement of production one new central production shaft, one new Decline in the western side of the property, one new ventilation shaft, new sets of mining equipments and other infrastructure & equipments are proposed to be added to the present system. There will be a need of extra manpower of about 550 persons who will be hired from the nearby villages which are mostly tribal dominated.

Mining Method

The ore body occurs as six different sets of lodes. Based on their positions, the ore-bodies are also termed as Lode-1, Lode-2, Lode-3, Lode-4, FWA lode and FWB lode. The stripping ratio, even for the upper horizon of ore body works out to be more than 1:20. Thus to extract it economically, the underground method of mining has been found logically to be more suitable for the deposit. Based on the existing nature of the deposit and the prevailing geological conditions, Horizontal Cut and Fill with Post Pillar mining method has been adopted for mining of the deposit.

The Horizontal Cut and fill (HCF) is a method to excavate one slice of ore and filling back the same by waste rock /mill tailing etc. This fill forms the platform for men and machinery to work on it to excavate the next slice. At Turamdih, slices will be taken from the lower horizon to the upper horizon.

The sequence of operation is as given below.

- a. **Exploratory Drive following the Footwall Contact:** First of all ore drive of size 5.0 m x 3.2 m approximately is developed along the footwall contact from one end of the proposed stope block to other end along the strike. Often each stope is distinct ore lens and the above development of ore drive establishes the length and behavior of the ore body at that level. Ore drives at the upper and the lower levels are developed in similar manner.
- b. **Exploring the Hangwall Contact:** The ore drive as developed at the lower level following the footwall contact is widened to expose the hang wall subject to a maximum width of 8.0 m. Beyond this width, regular 5 m x 5 m pillars are left systematically in both strike and dip directions up to the hangwall contact. This establishes the exact horizontal width of the ore body of the proposed stope block. This is especially important at the Turamdih as the ore body is lenticular and highly irregular in nature.

- c. **Return Air Route from the Stope:** At both extremities of the proposed stoping block, raises are put up to connect the lower level to the upper level for the purpose of establishing ventilation route.
- d. **Access Route for Trackless Equipment:** A ramp is developed in footwall rock to provide access into the stope for the track less equipments like Jumbos & LHDs. The ramp is developed either from upper level to lower level or from lower level to upper level or a combination of the two.
- e. **Footwall Drift:** The ramp is developed from a Footwall Drift developed parallel to the footwall drive by maintaining the required distance from the drive and serves as the main access route for men, machinery and materials.
- f. **Commencement of Slicing Operation:** The back of the ore drive at the lower horizon is stripped up to a height of 5.0 m followed by filling up to a height of 3.0 m from the stope floor. During slicing operation, physics and geology personnel provide necessary guidance about the continuity of the ore body and its vertical geometry with the help of radiometric probes. In this manner, the stoping cycle viz. slicing and filling continues from the lower level to the upper level.
- g. **Supporting:** The supporting of the stopes and other workings is done as per the approved SSR (Systematic Supporting Rule) approved by DGMS (Directorate General of Mines Safety) based on the rock mechanics study carried out by CIMFR (Central Institute of Mining & Fuel Research, Dhanbad).
- h. **Ventilation:** Proper ventilation of the stope is maintained by establishing exhaust air routes by developing raises to the upper level workings at regular intervals.
- i. **Illumination:** All the trackless equipment have their own illumination arrangement. Apart from this cluster lights are provided at places where supporting and other activities are carried out. The roadways are illuminated by CFL lamps.
- j. **Pollution Control:** The trackless equipment deployed in stopes are of international standard and have adequate pollution control systems. The workmen are provided PPEs. Water hydrants are provided for sprinkling water over the muck pile after blasting and before loading.
- k. **Underground Equipment:** The following trackless underground equipment are deployed for production and related services.
 - 1) **Single Boom Drilling Jumbos** (Boomer /Rocket Boomer of make Atlas Copco & Axera of make Sandvik) are used for drilling in development and stoping faces. (Hole depth = 3.4 m, hole diameter = 48 mm).
 - 2) A **rock bolting machine** (Boltec: Make Atlas Copco) is used for rock bolting operations in stopes and development districts. This is the first such equipment in use in underground in India.

- 3) Diesel LHDs (Scoop Trams: Make Atlas Copco, 3.5 Cu Yard) are used for muck clearing and loading operations.
- 4) Mine Trucks (MT: Make Atlas Copco, LPDT: Make Sandvik) are used to tram the ore from the development and stope faces to surface as well as for tramming waste rock from development faces to stopes for filling/surface for disposal.
- 5) Passenger Carriers (Make Normet /Sandvik) are used for transporting personnel from surface to working places.
- 6) Scissor Lifts are used as platform for persons to work at heights.
- 7) Crane Truck is used for various lifting and loading works in underground as well as in the workshops.
- 8) Electro-hydraulic Diamond Drilling Machine (Diamec): This Atlas Copco make diamond drilling machine is used for extensive underground exploration work.
- 9) Drilling Jumbo for Extension Hole Drilling: One of the drilling Jumbos is used for extension hole drilling up to the length of 15 m for exploration activities.

Hoisting System

Presently, hoisting of ore is being done using existing 8 degree decline and the vertical shaft of 5 m finished diameter. The present system is not adequate to handle 2500 tpd of ore from underground.

Designing, Sinking , lining, equipping and commissioning of a production shaft of 5 meter finished diameter along with suitable ore hoisting arrangement , a friction winder & unloading arrangement etc and a 8 degree decline are required to handle the extra production of 1000 tpd over the present level of production capacity of 1500 tpd. These two mine entries shall be in suitable place preferably at the central location of the property. The ore produced from different levels (viz. 100 m level, 140 m level & 180 m level) will be dumped to the crushing station (200 m level) through the central ore pass (100 m level /2nd level) to 200 m level (5th level /Crushing station). The crushed ore will be stored in an underground bin of 1000 tonne capacity below the crushing station and will be subsequently fed to a belt conveyor located at the hoisting Station (230 m level /6th level). The conveyor will feed the ore to a pair of measuring pockets which will discharge it into the skips for hoisting it to surface. From surface, the ore will be stored in a bunker of 1000 ton capacity and will be subsequently transported by conveyor belt to the ore processing plant. The ore produced from the upper levels will be brought to surface directly by the underground mine trucks and will be transferred to the ore processing plant ground hopper located at a distance of about 100 metre from the 8-degree Declines.

Blasting

Burn cut pattern will be used for blasting in drifts, drives and cross cuts. Small diameter emulsion explosives with delay electric detonators, both millisecond Delay and Half Second Delays will be used. This explosive is suitable for dry as well as wet holes. The explosive consumption per face of 5 m x 3.4 m x 3.2 m size is estimated to be 135 kg, whereas the rock broken at 96% pull will be around 130 t. Thus powder factor can be achieved as 0.9 t/kg. It is anticipated that in spite of all precautions, few boulders and few humps may be found during the development and stoping operations. These boulders and humps will require secondary blasting. Secondary blasting will be carried out either in the form of pop shooting or by regular jack hammer drilling as the case may be. A small site magazine with storage capacity of 1.5 tonne (1500 kg) and 5000 nos. detonators has been set up. Apart from this, there exists one 44 tonne capacity magazine also with the capacity of storing 1,76,000 nos. of detonators for UCIL group of mines at Jharia. The site magazine caters the day to day need of the Turamdih uranium mine.

Mine Ventilation System

Presently the mine has two intake airways Decline and Main shaft. There are two return airways (Western Ventilation Shaft & Central Ventilation Shaft). Two number of VF - 3000 fans are working in the return airways with an exhaust capacity of 350 CuM /sec. This system will be requiring changes with installation of a new ventilation shaft at the western extreme of the ore body. This additional fan shall fetch another 175 CuM /sec to the mine which will suffice the requirement of ventilation of the mine with the rated capacity of 2500 tpd.

The modified ventilation system shall have four numbers of mine entries two numbers of Shaft and two numbers of Declines.

Major Mining Equipments in Mine

List of equipment being used in the Turamdih mine along with their power consumption are given in **Table** .

List of Major Equipments in Turamdih Mine

Sl.	Name	No. in Operation	Total kW
Surface Installations			
1	Winder	02	800.0
2	Ventilation fan	3	1200
3	Pumps	09	250

Sl.	Name	No. in Operation	Total kW
4	Workshops including smithy and foundries	01	2
5	Diesel Generator	04	4 x 1 MVA
6	Compressors and accessories	4	1000
Underground Installations			
1	Ventilation fan (Booster)	02	150
2	Ventilation fan (Auxiliary)	20	746.0
3	Dewatering Pumps	08	492.0
4	Crusher	01	250.0
5	Drill Jumbo	08	480
6	Boltec-235	01	55.0
7	Mine Truck	10	2000.0
8	Scoop Tram	12	2000
9	Multi Purpose Carrier	03	195.0
10	Passenger Carrier	03	220
Mobile Equipment			
1	Pay Loader	03	280
2	Escort Crane	01	31.5
3	Dumper	02	23.5
4	Bus	03	50
5	Jeep	02	82.0
6	Explosive Van	02	94
7	Fork Lift Truck	01	48.0
8	Dozer	01	239.0
9	Truck	01	23.5

3.5.4 Mine De-watering

There are two main pump chambers at 140 m level (3rd level) and 60 m level (1st level) with capacity of 300 M³ per hour. Apart from this, there will be smaller capacity pump chamber at the shaft bottom level [260 m level) with a pair of 25 hp, 90 Cu M /hr roto pumps. There is provision of emergency power for all the pump chambers. There is an elaborate network of drains to route the water coming from

different districts (stowing, drilling & seepage water) of each level to the main pump chambers.

3.5.5 Mineral Processing

ROM ore is crushed to -750mm size in underground is sent to hopper through conveyor belts. The ore size is further reduced to to (-) 25 mm. size in the Crushing Plant in three stages. Primary Crushing is done by Jaw Crusher to – 150 mm. size and after screening in two Single Deck Screens, –25 mm fractions is taken for grinding. Rest are further crushed in Cone Crushers & these are in close circuit with the single deck screens (25 mm. size). The uranium present in the ore is extracted through a hydro-metallurgical process as magnesium di-uranate / uranium peroxide. This product is dispatched to Nuclear Fuel Complex, Hyderabad for further processing.

3.6 RAW MATERIALS

The only raw materials required for the project are explosives for blasting ore and HSD for fuelling mining and allied machinery and vehicles.

The present explosive consumption is 0.5 tons per day, which is expected to increase to 1.5 tons per day. The present HSD consumption is 0.57 tons per day which is expected to increase to 1.4 tons per day.

3.7 RESOURCE OPTIMIZATION / RECYCLING AND RESOURCE

3.8 SITE SERVICES

3.8.1 Water Requirement:

Water requirement will be 3430 m³/day out of which 3170 m³/day will be met by mine discharge water and 260 m³/day will be tapped from the existing water supply system.

3.8.2 Power Requirement

The present power consumption is 2.0 MW. The power demand is not expected to increase to the level of 5 MW. The power is supplied by Jharkhand State Electricity Board (JSEB) through its main sub-station at Ichra sub- station. Two nos. 1000 KVA DG sets (1 working, 1 standby) have been installed to meet the emergency power requirement. Another two numbers of 1000 KVA DG sets will be required for the enhanced production capacity.

3.8.3 Amenities

Turamdih mine is an operating mine with a sound infrastructure in place. The following facilities have been created and no additional facility is required for the proposed expansion project.

1. Compressor House and Compressors
2. Electric Sub Station and Power distribution net work
3. Surface workshops
4. Main Ventilation system
5. Explosive Magazine
6. Lamp Room
7. Time Office
8. Canteen
9. Pit head baths & Locker Room for workers
10. Office complex

3.9 WASTES

About 150000 t of waste rock is will be generated during excavation of new decline and shaft and out of 150000 ton waste, about 100000 ton of the waste rock will be brought to the surface and kept in the designated waste dumping area within the lease area. The rest of the waste shall be used underground for stowing purpose in the excavated voids of stope. During actual operation stage of expanded mine about 150000 ton waste will be generated in the process of mine development and this waste shall be used as filling material at excavated voids in stope.

4.0 SITE ANALYSIS

4.1 CONNECTIVITY

Turamdih mine is located at a distance of about 5 km by Road from Tatanagar Railway of Howrah-Mumbai BG line of SE Railways. At present the road is adequate to handle the traffic. The nearest functional airport is Sonari at Jamshedpur which is about 20 km NW of the mine.

4.2 LAND FORM, LAND USE, OWNERSHIP

The existing mining lease area of Turamdih Mine is 225.487 ha (557.18 ac) under villages of Nandup, Banduhurang and Purihasa. The leasehold area consists of 35.415 ha forest land, 53.035 ha government land and 137.037 ha private land. Additional forest area of 12.857 ha will be added in existing mining lease.

4.3 TOPOGRAPHY

Turamdih mine is an underground mine and hence alteration of the surface topography is not envisaged.

4.4 LAND USE

Turamdih underground Project includes three villages namely Nandup, Purihasa and Banduhurang. The present lease area of Turamdih Mine is 225.487 ha. The details of present land use of mining lease are shown in following table. The major portion of the lease area is covered under open land category which will be converted into green belt in future. Besides this the other land use classes occurring in the lease are agricultural land, industrial infrastructure, green belt and forest land along with settlement. The lease area doesn't have any major water body.

Land Use Pattern of Lease Area

Land Use Category	Present Lease area in ha
1. Agricultural land	17.80
2. Forest land	35.42
3. Open land	83.397
4. Waste land	3.22
5. Green belt/ Plantation	31.00
6. Industrial Establishment	38.87
7. Infrastructure and colony	8.78
8. Settlement	1.80
9. Roads	5.20
Present Lease Area in Total	225.487
Additional lease (forest area)	12.857
Proposed Lease Area in Total	238.344

4.5 EXISTING INFRASTRUCTURE

Refer Clause 3.8.3 above

4.6 SOIL CLASSIFICATION

As per the District Planning Map of Purbi Singhbhum, published by National Atlas and Thematic Mapping Organization, Kolkata the soil of the area where Turamdih Mine is located is classified as "Red Loamy Soil".

4.7 CLIMATE

The study area lies in tropical region where climate is characterized by very hot summers and cool winters. The nearest observatory of India Meteorological Department (IMD) is at Kadma Colony in Jamshedpur, about 16 km away. Summer is typically from mid March to mid June when temperature ranges from a maximum of 40.1°C during day time to a minimum of 18.6°C at night. Winter is from December to

February when the maximum temperature during day goes up to 29.4°C and minimum temperature at night becomes 11.6°C. The average annual rainfall is 1321 mm. The South-west monsoon lasts from mid June to mid September and the area gets more than 80% of the annual rainfall during this period.

4.8 SOCIAL INFRASTRUCTURE AVAILABLE:

The mine is located in a rural area. The nearest town / city Jamshedpur is about 15 km from the mine, which has all necessary social infrastructures. The social infrastructure available in the mine township has been described under clause 3.8.3 above.

5.0 PLANNING BRIEF:

5.1 PLANNING CONCEPT:

The entire infrastructure necessary for the expanded mine are already in place. All the machineries are also available. The higher production will be achieved by increasing machinery utilization. The rated capacity of the expanded mine 0.75 million TPA shall be attained after receipt of necessary statutory clearances.

5.2 LAND USE PLANNING:

The existing land use in the mine lease has been given in clause 4.4 above. The land use shall remain unchanged during the operation of the expanded mine. When the reserves are exhausted, the mine will be shut down as per the Approved Mine Closure Plan. Some of the infrastructure will be dismantled. Others will be handed over to the State Government or the local village panchayats. The waste dumps will be stabilized and biological reclaimed. Plantations will be carried out on benches and floors of unfilled quarries.

5.3 ASSESSMENT OF INFRASTRUCTURE DEMAND

Most of the infrastructure such as Mine Office, surface material handling plant, electrical sub-station, stores, explosive magazine, workshop, weigh bridge, rest shelter, canteen, vocational training centre, medical unit etc. are already in place. Some of these such as surface material handling plant, explosive magazine and workshop will have to be expanded to cater to increased excavation and handling of ore.

5.4 AMENITIES / FACILITIES

All necessary amenities such as rest shelters, canteens, pit head baths, vocational training centre etc. already exist.

6.0 PROPOSED INFRASTRUCTURE:

New production shaft, Decline, Ventilation Shaft, workshop, Mine Vehicle, Crusher and related equipment, Pumping, Ventilation Fan, Winding Equipment etc

7.0 REHABILITATION & RESETTLEMENT (R&R) PLAN :

The mine is operating since 1995. Since the proposed expansion project does not envisage any leasing and / or acquisition of additional land no rehabilitation & resettlement is required for this project.

8.0 PROJECT SCHEDULE & COST ESTIMATE

8.1 LIKELY DATE OF START OF CONSTRUCTION AND LIKELY DATE OF COMPLETION:

Project is likely to be completed in three years i.e. by 10.10.2020.

8.2 ESTIMATED PROJECT COST ALONG AND ECONOMIC VIABILITY OF THE PROJECT

The estimated cost of the project is Rs 300 crore.

9.0 ANALYSIS OF PROPOSAL (FINAL RECOMMENDATION)

The proposed expansion project will have the following benefits:

- Improve supply of indigenous uranium ore and thereby increase supply of fuel for India's nuclear programme.
- Generate direct as well as indirect employment.
- Pave way for further peripheral development of a predominantly tribal area.